

SYMPOSIUM 2.0 WEBSITE

Software Requirements Specification (SRS)

Version 2.1 (V2 Build) — Consolidated & Updated

Department of Computer Science & Engineering
Shridevi Institute of Engineering and Technology (SIET)
Hosting Club: Creative Codex Club

Status: V2 build complete — ready for team/faculty review

Document updated to reflect the actual final V2 build, including all deviations from and additions beyond the original V2 Corrected specification.

Document Information

Field	Value
Project Name	Symposium 2.0 Website
Organisation	Department of Computer Science & Engineering
Institution	Shridevi Institute of Engineering and Technology (SIET)
Hosting Club	Creative Codex Club
Document Type	Software Requirements Specification (Master Knowledge Base)
Version	2.1 — reflects the completed V2 build
Status	V2 build complete, verified, ready for team/faculty review before generation-to-production

Note on this revision: This document updates the original SRS v2 (Corrected & Consolidated) to reflect the actual V2 build as implemented — including the shift from an external Google Form to a full on-site dynamic registration module, a revised visual design system, several new pages and sections not in the original scope, and corrected database/backend logic discovered and fixed during development. Every deviation is marked inline, in the same convention as the original document’s own “Fix applied” notes.

V2 Change Summary

This chapter is new in V2.1. It exists so any reader — team member, faculty reviewer, or a future maintainer — can understand exactly what changed from the original SRS v2 (Corrected) without having to compare documents line by line. Every item below is reflected in the corresponding chapter later in this document.

1. Registration: External Form → Full On-Site Module

NEW: The original SRS specified that "the site does not process registrations itself in v1 — it links out to an external platform" (Chapter 6.4). This has been superseded. V2 implements a complete, on-site, database-driven registration system, replacing the Google Form entirely.

Reasoning: the team judged that an on-site system looks more professional for a national-level event and gives the department a durable, reusable asset across future editions, while remaining low-risk because it stayed deliberately simple (no drag-and-drop form builder, no template library — see Chapter 6 for the full design).

2. New Pages Beyond the Original Sitemap

NEW pages, not in the original SRS: Travel & Accommodation, Judges/Mentors (revealed live during the event, not published in advance), Testimonials, Code of Conduct, Theme/Problem-Statement Reveal (flexible — works whether the event ends up using fixed problem statements or open ideas).

These were added because they materially strengthen credibility and trust for a national-level, first-time-feeling event — see Chapter 2.4 (Objectives) for how they connect back to the project’s own stated goals.

3. Visual Design System Replaced

CHANGED: Chapter 5’s colour palette and component styling are replaced. The original cyan-accent, rounded-pill system is replaced with a flatter, more restrained “corporate” system (solid navy + solid blue, sharper radii, no gradients or glow) — chosen deliberately after comparing it against a more decorative glass/gradient alternative, on the reasoning that restraint reads as more trustworthy for this audience. See Chapter 5 for the full updated system.

4. Database Schema Extended

NEW tables added in V2: `event_registration_settings`, `registration_fields`, `registrations`, `registration_answers`, `team_members`, `judges`, `testimonials`, `travel_info_sections`, `code_of_conduct`, `theme_reveal`. The original edition-scoped `registration_settings` table is kept for backward compatibility but is no longer written to by V2 code. See Chapter 9.

5. Two Real Backend Bugs Found and Fixed During the Build

FIX APPLIED: The generic content API only ever filtered results by `edition_id`; any table scoped by `event_id` instead (registration fields, per-event registration settings) silently ignored that filter and returned every event’s rows mixed together. Fixed to support both filter types, verified against live multi-event data.

FIX APPLIED: During that same fix, two tables (`downloads`, `live_updates`) were incorrectly assumed to have a `display_order` column they don’t actually have, which would have crashed those two endpoints whenever filtered. Caught by checking every table programmatically against the schema rather than from memory, and corrected.

6. Security Additions Beyond the Original Scope

NEW: rate limiting on the admin login route and the public registration endpoint (prevents brute-force and spam submission); a honeypot field on the registration form (bot protection without a third-party CAPTCHA dependency); server-side file type and size validation on payment screenshot uploads; CORS locked to a configurable allow-list instead of left open.

7. Navigation Restructured

CHANGED: Chapter 3.4’s navigation hierarchy is implemented with grouped dropdowns (About / Events / People / Media / Resources) rather than the flatter structure used in the original V1 build, to keep the bar clean as the page count grew from 8 to 21. See Chapter 3 for the updated hierarchy.

8. Footer: WhatsApp Link Added

NEW: the footer social row now includes a real WhatsApp community link (previously Instagram/LinkedIn/YouTube only), per the team’s decision to prioritise WhatsApp as the primary community/query channel over a query-handling backend.

9. Admin Access

NEW: a deliberately unobtrusive “Admin” text link is included in the footer copyright line on every page, rather than a prominent nav button — balances discoverability for legitimate admins against not inviting casual probing of the login form.

Chapter 1-2 — Project Overview, Vision & Objectives

Unchanged from SRS v2 Corrected. The background, problem statement, proposed solution, vision, mission, core objectives, success criteria, stakeholders, target audience, scope, assumptions, and constraints all remain exactly as originally specified. The V2 build fulfils and extends this scope; it does not revise it. See the original SRS v2 Corrected document for the full text of these chapters.

One clarification worth noting here: Objective 4 (“Digital Branding”) and Objective 2 (“Centralised Information”) are the direct justification for several of the new pages added in V2 (Testimonials, Judges, Travel & Accommodation) — these were added specifically because they serve objectives already stated in the original document, not as unrelated scope creep.

Chapter 3 — Information Architecture, Sitemap & Navigation (Updated)

The IA principles (3.1, 3.9, 3.11) are unchanged. The sitemap, navigation hierarchy, and URL structure are updated below to reflect the V2 build.

3.2/3.3 Updated Sitemap

All original pages remain. New pages added in V2:

- Individual Event Detail (was previously described but not built as a real, wired page)
- Downloads / Resource Centre (built and live-wired)
- Members & Designations (built and live-wired)
- Achievements & Records (built and live-wired)
- Sponsors & Partners (built and live-wired)
- Symposium Recap (built and live-wired)
- Live Updates (built, live-wired, with teasers on Home and Events pages)
- Registration — now a real per-event status hub, not a single external link (see Chapter 6)
- Registration Form (new — the actual dynamic, per-event form)
- Travel & Accommodation (new)
- Code of Conduct (new)
- Privacy Policy (new)
- System pages: 404, Coming Soon, Thank You (registration receipt), Registration Closed, Registration Full

3.4 Updated Navigation Hierarchy

Nav Item	Dropdown Contents
Home	—
About ▼	The Symposium, SIET, Dept. of CSE, Creative Codex
Events ▼	All Events, Schedule, FAQ
Register	Standalone CTA button, not a dropdown
People ▼	Organising Committee, Members & Designations
Media ▼	Gallery, Symposium Recap, Previous Editions, Achievements
Resources ▼	Downloads, Travel & Accommodation

Footer-only (not in main nav, per the “uncluttered nav” principle): Sponsors, Live Updates (also teased contextually on Home and Events), Code of Conduct, Privacy Policy, Contact, and a deliberately unobtrusive Admin link in the copyright line.

3.10 Updated URL Structure

/event-detail?id= , /downloads , /members , /achievements , /sponsors , /recap , /live-updates , /registration , /registration-form?event= , /travel-stay , /code-of-conduct , /privacy-policy , /thank-you?id= , /404 , /coming-soon , /registration-closed , /registration-full — added to the original list from Chapter 3.10.

Chapter 4 — Page-by-Page Functional Requirements (Updated)

Pages 1–20 from the original SRS remain valid as written, with one correction: Page 8 (Registration) is fully superseded — see below. New pages follow.

Page 8 — Registration (Superseded)

FIX APPLIED: The original spec for this page assumed a single edition-wide external registration link. It is replaced entirely by an on-site, per-event registration hub and dynamic form — see Chapter 6 for the full functional specification.

New Page — Individual Event Detail

Full description, rules, eligibility, a Judges/Mentors section (empty/teaser until the admin flips `is_revealed` to true for each judge, consistent with the “revealed live during the event” decision), rounds/timeline, sidebar with team size, venue, prize, coordinator, and a Register button that links to that specific event’s registration form.

New Page — Downloads / Resource Centre

Real files fetched live from the database, auto-sorted into Event/Participant/Post-Event sections by keyword matching on title (the schema does not have a dedicated category column; this was a deliberate lower-risk choice over a schema change this late in the build).

New Page — Members & Designations, Achievements, Sponsors, Symposium Recap

Each live-wired to its own database table, with graceful fallback to placeholder content when nothing has been published yet (e.g. Recap stays a placeholder until `published_at` is set by the admin).

New Page — Live Updates

Real-time, timestamped updates. Note: the actual implemented schema for this table is simpler than what Chapter 9 of the original SRS described (a single `update_text` field, not separate title/description/category fields) — this matches what the original V1 codebase actually built, and V2 chose not to change it further given the added risk of a schema migration this late; see Chapter 9.

New Pages — Travel & Accommodation, Code of Conduct, Privacy Policy

Admin-editable content pages, edition-scoped so archived editions retain their own historical wording (Code of Conduct is versioned per edition for exactly this reason).

New System Pages

404, Coming Soon, Thank You (registration receipt, fetches the real registration by ID and offers Print/Download), Registration Closed, Registration Full — all implemented per the original Page 20 concept, now actually built.

Chapter 5 — UI/UX Design System (Updated)

FIX APPLIED / CHANGED: This chapter’s colour palette and component treatment are replaced. The team compared two directions — a glass/gradient "modern tech" style and a flat "corporate/enterprise" style — and selected the latter as the safer, harder-to-get-wrong choice for a national-level institutional event.

5.4 Updated Colour Palette

Role	Colour
Primary dark (navy)	#0f1b33
Accent (solid blue, not gradient)	#0b5fff
Off-white background	#f9fafb
Slate (body text)	#47566b
Hairline borders	#e4e7ec

Explicitly avoided, by deliberate decision: gradients, glassmorphism/blur, glow effects, and a wide multi-colour accent palette — all present in the alternative concept that was considered and rejected.

5.5 Typography — Unchanged

Poppins (headings), Inter (body), Orbitron (countdown numerals only) — retained from the original specification.

5.12 Animation — Extended

In addition to the original list (fade-in, slide-up, hover elevation, number counters), V2 adds: a slim top-of-page loading bar while live content fetches from the backend (fades cleanly on completion or error, never gets stuck), and a restrained pulsing dot on the Live Updates teaser (respects `prefers-reduced-motion` , consistent with the original accessibility principle in 5.14).

Chapter 6 — Functional Requirements (Updated: Registration Module)

Sections 6.2, 6.3, 6.5–6.17, 6.18, 6.19 are unchanged. Section 6.4 (Registration) is fully superseded by the specification below.

6.4 (Superseded) — On-Site Dynamic Registration Module

Design Principle

Fully dynamic and database-driven so that adding, removing, or changing events and their registration fields in future editions never requires a code change — while deliberately avoiding a full drag-and-drop form-builder UI (a Google-Forms-style feature considered and scoped out) in favour of a simpler add/edit/delete-one-at-a-time admin interface. Both achieve the same reusability goal; the simpler version was chosen to reduce risk given the team’s size and timeline.

Per-Event Configuration (admin-set, per event)

- Registration status: Not Set / Coming Soon / Open / Paused / Closed / Full
- Team size minimum and maximum
- Registration fee amount
- UPI QR code image (admin-uploaded, amount baked into the code by the faculty coordinator who generates it — payment goes directly to the faculty account, never to the website or any student-held account)
- Seat capacity, with live registered-count tracked automatically from real submissions (never manually editable, to preserve data integrity)
- Optional registration deadline

Dynamic Fields (admin-defined, per event)

Supported field types: Short Text, Long Text, Email, Phone, Number, Dropdown, Radio, Checkbox, Date, URL, File Upload. Each field has a label, type, required/optional flag, and display order (set by the order fields are added — no drag-to-reorder). The team’s intended default field set for this edition (to be added by the admin when configuring each event) is: Full Name, Email, Phone, State, District, College Name, Department, Year of Study, USN (mandatory, format intentionally unvalidated since university formats vary), LinkedIn/GitHub profile (optional), How did you hear about us, Agree to Code of Conduct (checkbox), plus Team Name and per-member details for team-based events.

Submission Flow

1. Participant selects an event from the Registration hub (only shown as registerable if status is Open, deadline not passed, and capacity not reached)
2. Fills the event’s dynamic fields, plus team member details if applicable
3. Uploads a UPI payment screenshot (JPG/PNG, max 5MB, validated server-side by both type and size) and enters the UTR number
4. System validates all required fields, team size bounds, and a hidden honeypot field (bot protection, chosen over a third-party CAPTCHA to avoid an external dependency)
5. On success: a unique Registration ID is generated only after successful database commit (format: `SYM<edition>-<EVENTCODE>-<sequence>`), seat count increments atomically, and the participant is shown a printable/downloadable receipt
6. On any failure: the entire transaction rolls back — no partial registration is ever created

Duplicate Handling

Participants are not blocked from registering for multiple events. The admin dashboard flags possible duplicates (matching email, phone, or USN across registrations for the same event) for manual review — never auto-blocked.

Admin Capabilities

Search/filter registrations, mark payment status (Verification Pending → Paid), export event-wise to Excel (.xlsx) or CSV ordered by Registration ID, view flagged duplicates.

Explicitly Out of Scope (by deliberate decision)

No payment gateway integration (manual UPI + screenshot only), no email confirmation (on-screen receipt only), no participant login or self-service editing after submission (one-shot; corrections go through the admin), no drag-and-drop field builder, no reusable form-template library.

Chapter 7 — Non-Functional Requirements (Extended)

All original NFRs (7.1–7.13) remain in force. V2 adds the following, discovered as genuinely necessary during the build rather than specified upfront:

- **Rate limiting:** admin login capped at 8 attempts per 15 minutes per IP; public registration submission capped at 10 attempts per 10 minutes per IP.
- **CORS:** restricted to a configurable allow-list of known frontend/admin origins, not left open.
- **File upload security:** server-side MIME type and extension validation, 5MB size cap, randomised safe filenames on save (never the participant's original filename).
- **Error handling:** a central error handler ensures no internal stack traces or file paths are ever leaked to the client in any failure case — found and fixed during testing after an early CORS-rejection response leaked a full server path.
- **Dependency hygiene:** the workspace is kept at zero known vulnerabilities (verified via `npm audit`); one dependency (an Excel-export library) was swapped twice during development after each candidate was found to carry real, flagged vulnerabilities in its dependency chain.

Chapter 8 — Admin Dashboard (Updated)

All original modules (8.7–8.19) remain. V2 adds a dedicated **Registrations Hub** module, structured as four tabs per event:

- **Setup** — status, team size, fee, QR code, seat capacity, deadline
- **Form Fields** — add/delete dynamic fields (no drag-reorder, add-in-desired-order instead)
- **Submissions** — search, view, mark payment status, export Excel/CSV
- **Duplicates** — flagged possible-duplicate entries for manual review

V2 also adds management screens for every new content type: Judges/Mentors (including the reveal toggle), Testimonials, Travel & Accommodation sections, Code of Conduct text.

Chapter 9 — Database Design (Updated)

All original tables (9.5) remain unchanged in structure. The naming convention locked in 9.8 (plural snake_case tables, singular `_id` primary keys) was followed exactly for every new table — verified programmatically against the live schema, not assumed.

New Tables (V2)

Table	Key Columns
event_registration_settings	event_reg_id, event_id, status, team_size_min, team_size_max, fee_amount, qr_code_url, seats_total
registration_fields	field_id, event_id, field_label, field_type, field_options, is_required, display_order
registrations	registration_id, event_id, team_name, payment_status, payment_screenshot_url, utr_number, submitted_at
registration_answers	answer_id, registration_id, field_id, answer_value
team_members	team_member_id, registration_id, member_name, member_email, member_phone, member_usn
judges	judge_id, edition_id, event_id, full_name, designation, organisation, photo_url, is_revealed
testimonials	testimonial_id, edition_id, author_name, author_college, quote, photo_url, is_demo, display_order
travel_info_sections	section_id, edition_id, heading, content, display_order
code_of_conduct	conduct_id, edition_id, content, last_updated
theme_reveal	theme_reveal_id, event_id, is_revealed, reveal_content

FIX APPLIED: The original edition-scoped `registration_settings` table (external form URL, edition-wide open/closed flag) is kept in the schema for backward compatibility but is no longer written to by V2 code — registration status is now tracked per event via `event_registration_settings`, matching the shift to on-site, per-event registration.

FIX APPLIED: The `live_updates` table structure is simpler than what the original SRS 9.6 described (Title, Description, Category, Status fields). The actual implemented table — inherited unchanged from the original V1 build — has only `update_text` and `posted_at`. V2’s frontend and admin panel code were corrected to match this real structure rather than introducing a schema migration this late in the build.

Referential Integrity

All new tables were verified with real cascade-delete tests (deleting a parent event correctly removes its registrations, answers, and team members with zero orphaned rows) and a real end-to-end transaction test (a full registration submission, from field validation through payment record, committed correctly and rolled back cleanly on simulated failure).

Chapter 10 — Technology Stack & Architecture

Unchanged. HTML5/CSS3/vanilla JS frontend, Node.js + Express backend, MySQL database, three-layer architecture, npm workspace structure — all as originally specified in the SRS v2 Corrected. Two additions:

- **New backend dependencies:** `multer` (file upload handling), `write-excel-file` (Excel export — chosen after two earlier candidates, `exceljs` and `xlsx`, were both found to carry real flagged vulnerabilities in their dependency chains), `express-rate-limit`.
- **Deployment plan confirmed:** GitHub Pages (frontend), Render or Railway (backend + MySQL database), with the option of Firebase Storage specifically for payment screenshot files if the chosen host does not guarantee persistent file storage across red deploys.

Chapter 11 — Content Requirements & Asset Management

Unchanged in principle. The three-stage content process (Placeholder → Official → Post-Symposium) and placeholder conventions from the original Chapter 11 apply equally to all new V2 content types. One addition: real content for the About SIET and About CSE pages was sourced directly from the college’s own published website (shrideviengineering.org) during the V2 build, rather than left as generic placeholder text, given the team’s explicit request that institutional pages read as substantial and accurate rather than thin.

Chapter 12 — Roadmap & Status (Updated)

Phase	Status
Requirement Analysis (V1 + V2)	Complete
UI/UX Design (V2 — Concept B)	Complete
Database (V2.1 schema, 26 tables)	Complete, verified
Backend (core, security, registration module)	Complete, verified
Admin Panel (incl. Registrations Hub)	Complete, verified
Frontend (24 pages, all wired to live data)	Complete, verified
Full-project verification pass (32-phase, V2.1)	Complete
Documentation update (this document)	Complete
Content Integration (real events, dates, judges, sponsors)	Pending — team’s responsibility, ~2 months before the event
Deployment (GitHub Pages + Render/Railway)	Pending — not yet executed
Team/Faculty Review & Approval	Pending — required before this build is taken live

12.12 Updated Conclusion

The V2.1 build delivers everything specified in the original SRS v2 Corrected, plus a complete on-site registration system, several credibility-building pages beyond the original scope, a revised visual design system, and a number of real defects found and fixed through rigorous, repeated, multi-dimensional verification (functional, structural, and content-completeness) rather than assumed correct. Every deviation from the original specification is documented in this revision and its V2.1 changelog addendum, consistent with Principle 10 (AI Independence) — this document remains understandable and actionable by any developer or AI system without needing the original build conversation as context.